Relational Model

Relational Model

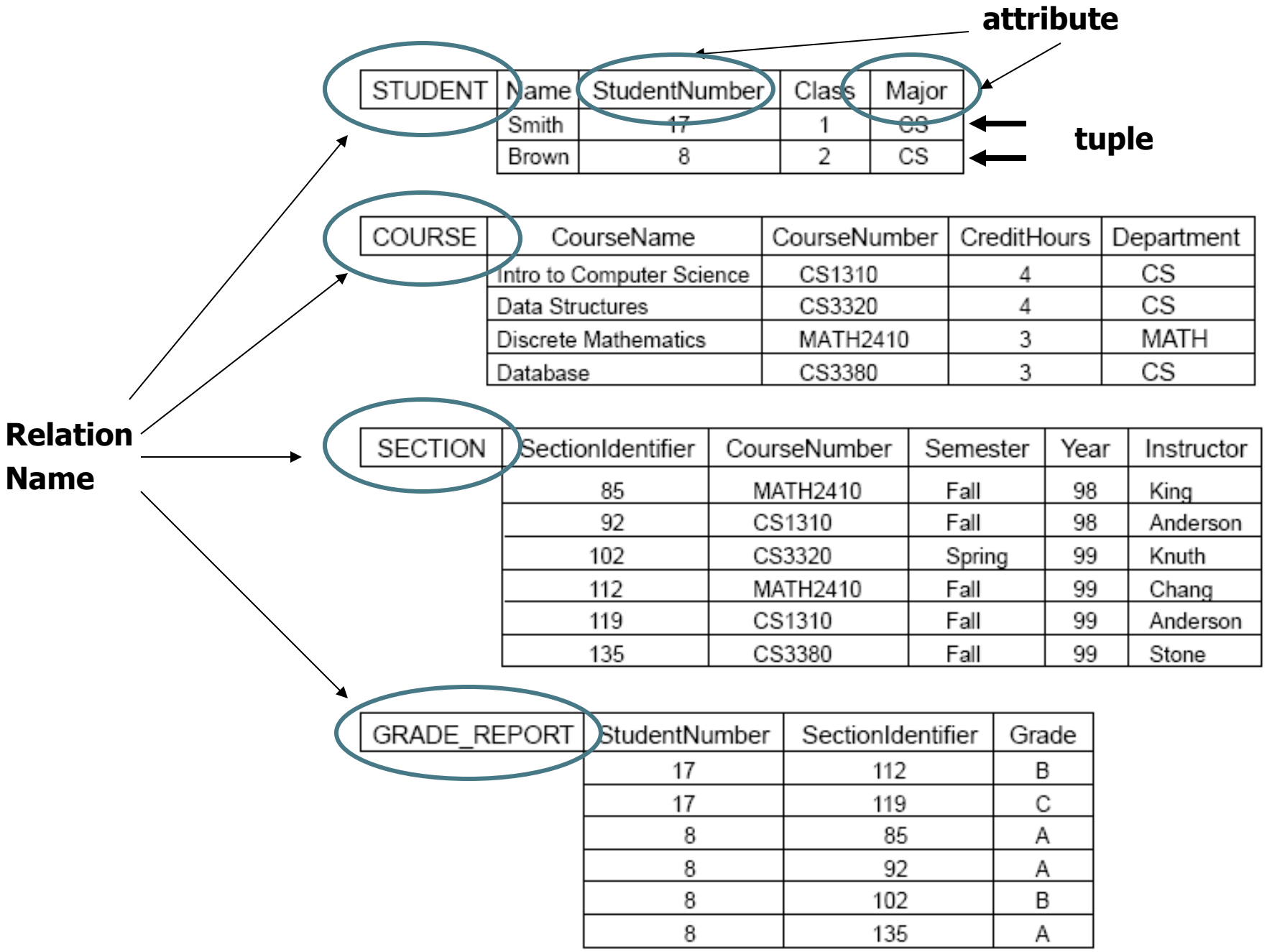
- Relational Model Concepts
- Relational Model Notation
- Relational Database Schema
- Domain & Key Constraints
- Referential Integrity Constraints
- Update Operations

Relational Model Concepts

- Relational Model
  - First introduced by Ted Codd (IBM) in 1970
  - Uses concepts of mathematical relation, set theory
  - First commercial implementation in early 1980s
  - Implemented on Oracle, SQL server, Access etc.

Relational Model Concepts

- Represent a DB as a collection of relations
- Represents a relation as a table:
- Table header as the name of the relation
- Columns as a set of attributes of the relation
- Rows as occurrences of the relation
- Row is called a tuple, column an attribute


attribute
tuple
Relation Name
STUDENT
Name
StudentNumber
Class
Major
Smith
17
1
CS
Brown
8
2
CS
COURSE
CourseName
CourseNumber
CreditHours
Department
Intro to Computer Science
CS1310
4
CS
Data Structures
CS3320
4
CS
Discrete Mathematics
MATH2410
3
MATH
Database
CS3380
3
CS
SECTION
SectionIdentifier
CourseNumber
Semester
Year
Instructor
85
MATH2410
Fall
98
King
92
CS1310
Fall
98
Anderson
102
CS3320
Spring
99
Knuth
112
MATH2410
Fall
99
Chang
119
CS1310
Fall
99
Anderson
135
CS3380
Fall
99
Stone
GRADE_REPORT
StudentNumber
SectionIdentifier
Grade
17
112
B
17
119
C
8
85
A
8
92
A
8
102
B
8
135
A

Relational Model Notation

- Domain of an Attribute
  - A domain D is a set of atomic (indivisible) values.
  - A domain may include a name, data type, and format, examples:
    - USA_phone_number: the set of ten-digit phone number valid in US.
    - GPA: Possible values of computed grade point averages; each must be a real number between 0 and 4
  - Domain of A_i is denoted by dom(A_i)
  - Data types: integer, char(20), date, float number, etc.

Relational Model Notation

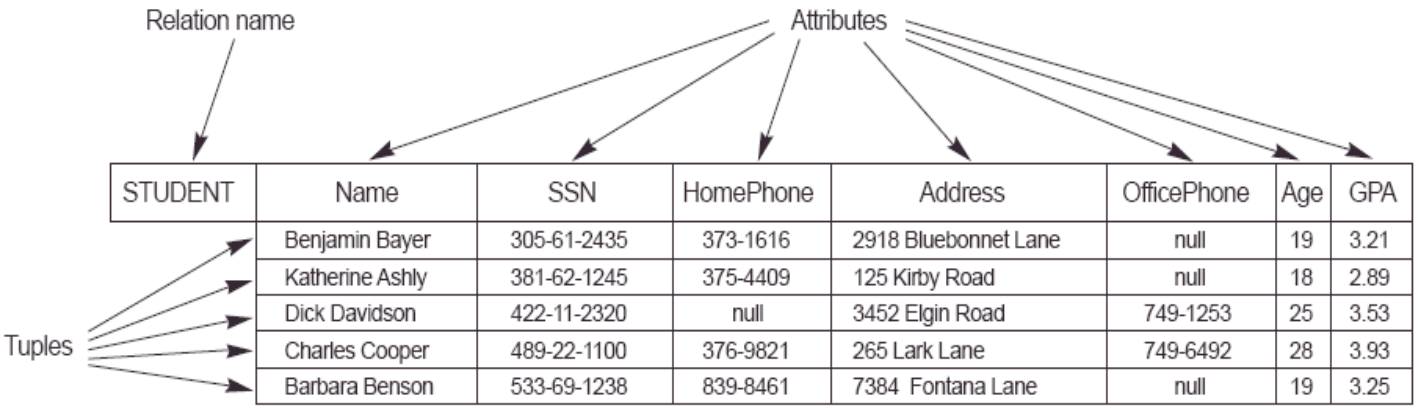
- Relation Schema R
  - Used to describe a relation
  - $R(A_1,\ldots,A_n)$
  - R: relation name
  - $A_1,\ldots,A_n$: a list of attributes
  - Degree of relation, n: number of attributes
  - i.e.) CAR(SerialNo, Make, Model, Year)
  - $R.A_i$ or $R[A_i]$: CAR.Make, CAR[Year]
  - Relation Intension: a definition of a relation

Relational Model Notation

- Relation State r(R) (= r)
  - Relation Extension: current status of a relation R as a set of n tuples
  - $r(R) = \{t_1,\ldots,t_m\}$
  - n-tuple $t = \langle v_1,\ldots,v_n\rangle$: ordered list of n values
  - Each V_i $(1<i<n)$ is an element from $dom(A_i)$
  - v_i (i^{th} value in tuple t) = $t.A_i$ (or $t[A_i]$)
  - ex) t.Make, t[Year]

Relational Model Notation

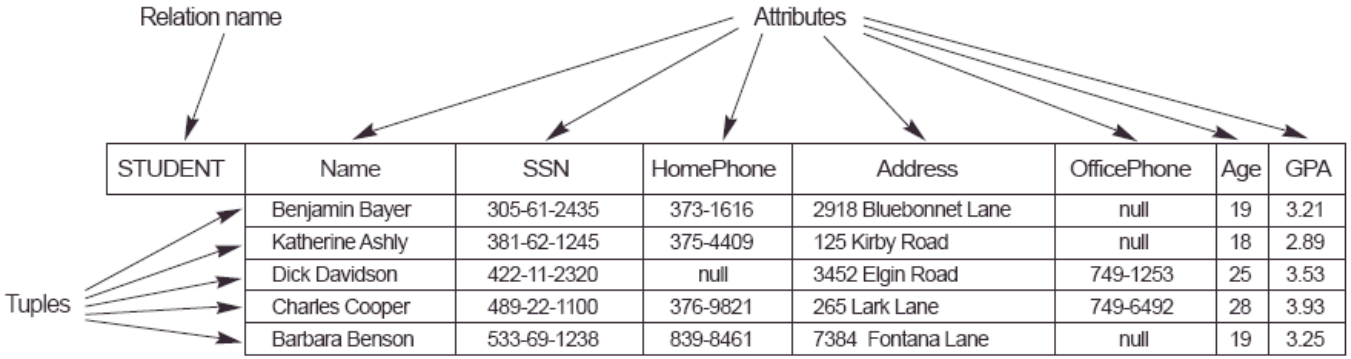
- A relation r(R) is a mathematical relation of degree n on the domains dom(A1), dom(A2)…dom(An)
- A relation r(R) is a subset of the Cartesian product of the domains that define R:
  - $r(R) \subseteq (dom(A_1) \times dom(A_2) \times .. \times dom(A_n))$
  - Current relation state reflects only the valid tuples that represent a particular state of the real world.



STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

Characteristics of a Relation

- Tuple Ordering
  - is insignificant (not part of a relation)
  - Both are identical relations



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Characteristics of a Relation

- Attribute Ordering
  - is significant if a tuple is considered as a set of values; t = $<v_1,\ldots,v_n>$
  - is insignificant if a tuple is considered as a set of (attribute,value) pairs; t = $<(A_1,v_1), (A_2,v_2), \ldots, (A_n,v_n)>$

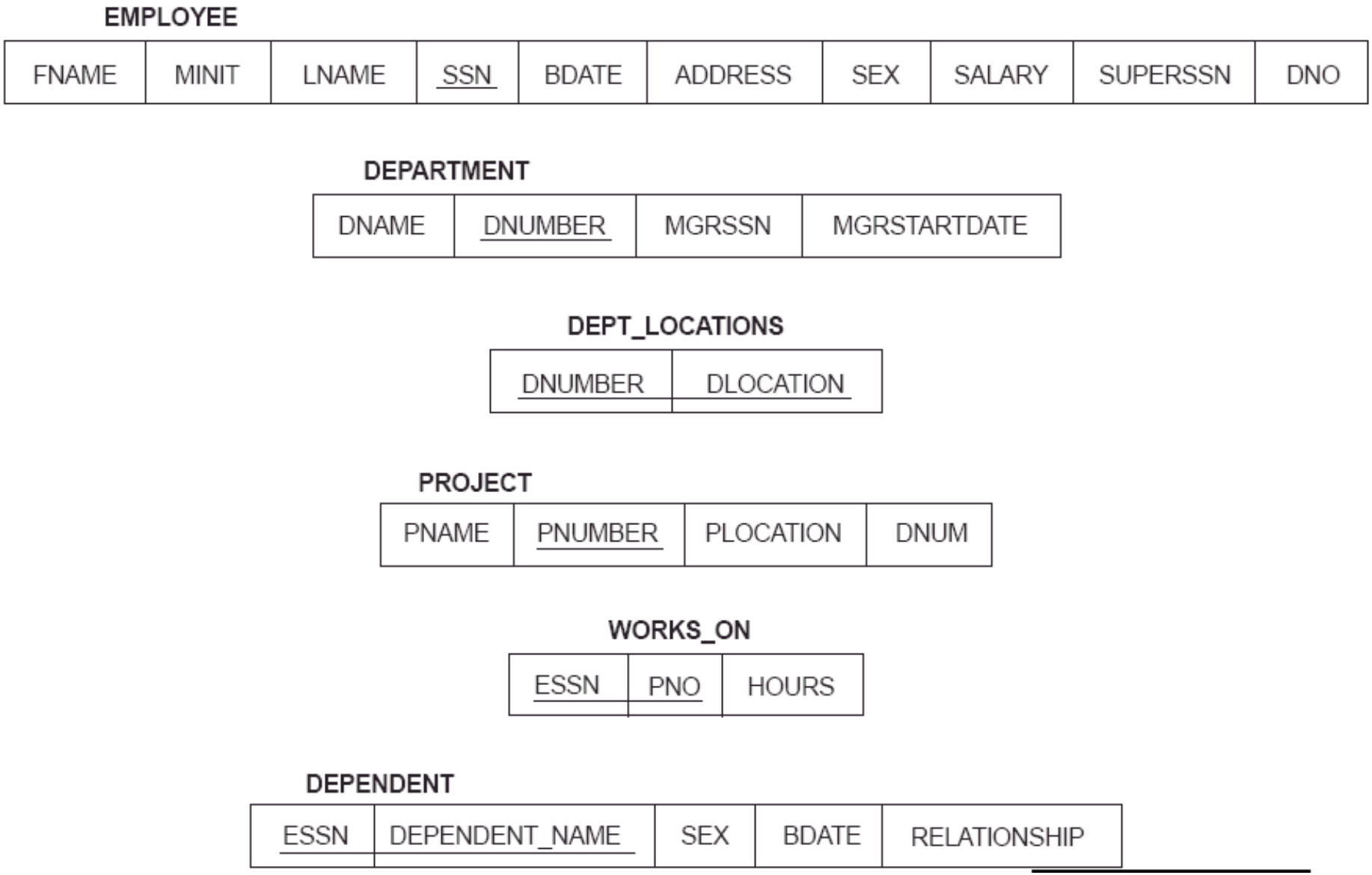
t = < (Name, Dick Davidson),(SSN, 422-11-2320),(HomePhone, null),(Address, 3452 Elgin Road), (OfficePhone, 749-1253),(Age, 25),(GPA, 3.53)>

t = < (Address, 3452 Elgin Road),(Name, Dick Davidson),(SSN, 422-11-2320),(Age, 25), (OfficePhone,749-1253),(GPA, 3.53),(HomePhone, null)>

Relational Database Schema

- Relational Database Schema
  - a relational database usually contains several relations
  - contains a set of relation schemas and a set of integrity constraints
  - $S = \{R_1, \ldots, R_n\}$

Schema diagram for the COMPANY relational database schema



Relational Database Schema

- Relational Database State of S
  - defined as a set of relation states at a particular point in time
  - DB = $\{r_1, \ldots, r_n\}$, where r_i is a state of R_i
  - r_i relation state satisfy the integrity constraints
  - relational constraints are expected to hold on every database state of a relational database schema

Relational Constraints

- Domain Constraints
  - value of each attribute A_i must be an atomic value from the domain dom(A_i)
  - atomic: not divisible into components
- Key Constraints: Superkey
  - for any distinct tuples t_1 and t_2 in r(R)
  - $t_1[SK] \neq t_2[SK]$ (SK is a subset of attributes)
  - SK is called a superkey of relation schema R

Relational Constraints

- Key Constraints: Candidate key
  - as a minimal superkey
  - a superkey such that removal of any attribute from the superkey ($SK - \{A_i\}$) results in a set of attributes that is not a superkey
  - a relation may have more than one key (candidate key)
  - a primary key (PK) is chosen from candidate keys to identify tuples in the relation

Relational Constraints

CAR	LicenseNumber	EngineSerialNumber	Make	Model	Year
	Texas ABC-739	A69352	Ford	Mustang	96
	Florida TVP-347	B43696	Oldsmobile	Cutlass	99
	New York MPO-22	X83554	Oldsmobile	Delta	95
	California 432-TFY	C43742	Mercedes	190-D	93
	California RSK-629	Y82935	Toyota	Camry	98
	Texas RSK-629	U028365	Jaguar	XJS	98

- **Superkey**
  - {LicenseNumber, Make}
  - {LicenseNumber, EngineSerialNumber, make, model}
  - {LicenseNumber, make, model}
- **Key (Candidate Key)**
  - {LicenseNumber}, {EngineSerialNumber}
- **Primary Key**
  - Select one candidate key: {LicenseNumber}

Relational Constraints

- Entity Integrity Constraints
  - Constraints on null: some attributes are not permitted to be null
  - No primary key value can be null because it is used to identify individual tuples in a relation
  - For any tuple t in r(R), t[PK] ≠ null
  - key constraints and entity constraints are specified on individual relations

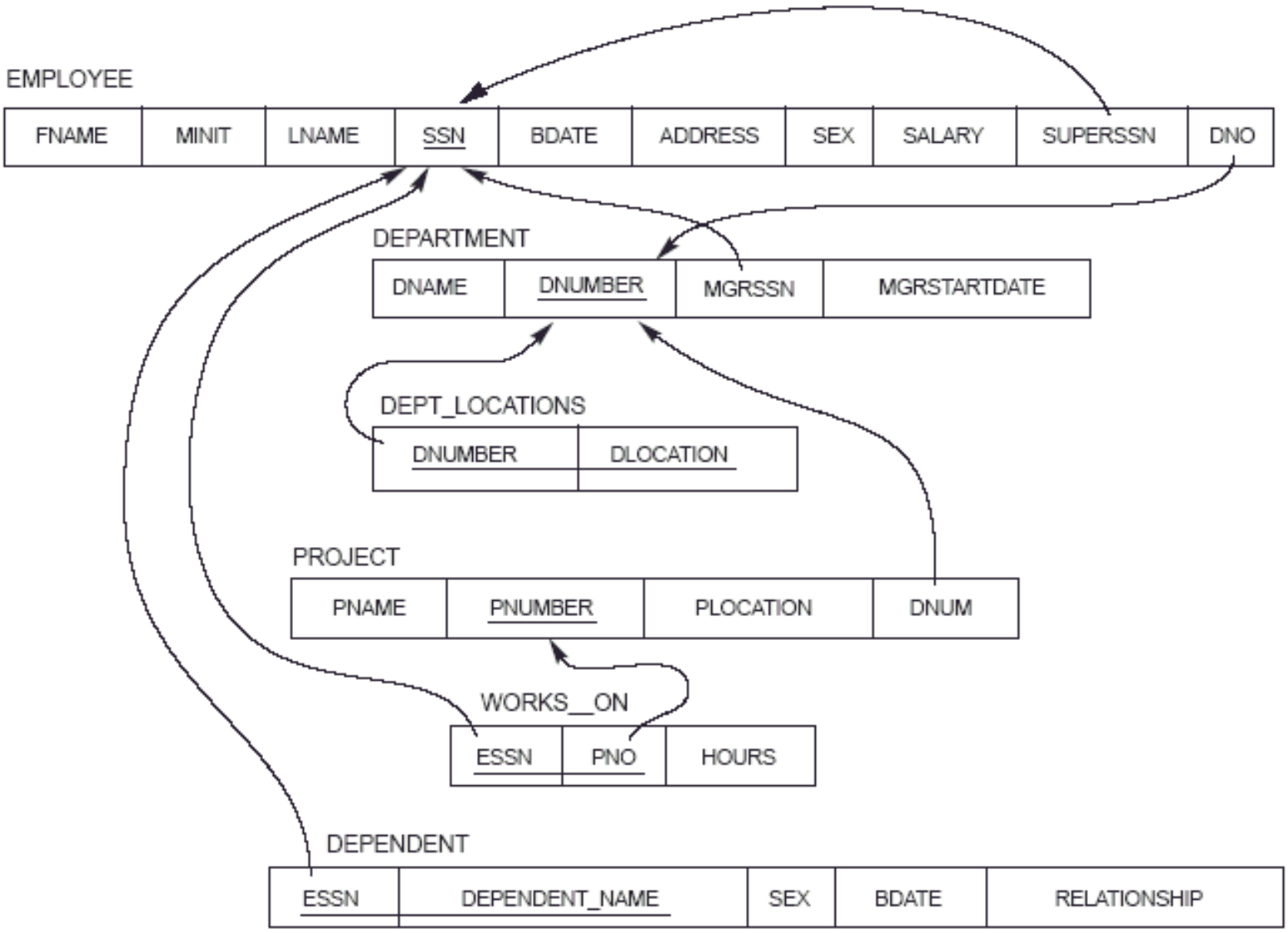
Relational Constraints

- Referential Integrity Constraints
  - specify a reference relationship between two relations.
  - used to maintain the consistency among tuples of the two relations
  - a tuple in one relation (R_1) that refers to another relation (R_2) must refer to an existing tuple in that relation
  - R_1:referencing relation, R_2:referenced relation

Relational Constraints

- Referential Integrity Constraints
  - tuples in the referencing relation R_1 have a foreign key (FK) that reference the primary key of the referenced relation R_2
  - A tuple t_1 in R_1 is said to reference a tuple t_2 in R_2 if $t_1[FK] = t_2[PK]$
  - values of a foreign key in R_1 are required to match values of the primary key in R_2 (from the same domain)

Figure 7.7 Referential integrity constraints displayed on the COMPANY relational database schema diagram.



Operations on Relations

- Update Operations
  - Insert, Delete, Modify
  - change the current status of the DB
  - should not violate constraints (domain, key, entity integrity, referential integrity)
- Retrieval Operations
  - Select, Project, Join, Union, Intersect, …
  - no change in the relation states (no violation)

Insert Operation

- Insert
  - insert a new tuple in a relation
  - can violate all four relational constraints
- Constraint Violations
  - Domain: if attrib value is not from the domain

    <Jay,Smith,999887777,60/04/05,M,28000,null,acct>
  - Key: if a key value already exists

    <Alice,Johnson,999887777,62/03/05,F,22000,null,5>

Insert Operation

- Insert <‘Cecilia’,’F’,’Kolonsky’,’null’,1960-04-05’,’6357 Windy Lane, Katy, TX’,F,2800,null,4> into Employee.

This insertion violates the entity integrity constraints (null for the primary key SSN), so it is rejected

Insert Operation

- Insert <‘Alicia’,’j’,’Zelaya’,’999887777’,1960-04-05’,’6357 Windy Lane, Katy, TX’,F,2800,’987654321’,4> into Employee.

- This insertion violates the key constraint because another tuple with the same SSN value already exists in the Employee relation, and so it is rejected.

Insert Operation

- Insert <‘Cecilia’,’F’,’Kolonsky’,’677678989’,1960-04-05’,’6357 Windy Lane, Katy, TX’,F,2800,’987654321’,7> into Employee.

- This insertion violates the referential integrity constraint specified on DNO because no Department tuple exists with DNUMBER=7.

Insert Operation

- Insert <‘Cecilia’,’F’,’Kolonsky’,’677678989’,1960-04-05’,’6357 Windy Lane, Katy, TX’,F,2800,null,4> into Employee.

This insertion satisfies all constraints, so it is acceptable.

Insert Operation

- Constraint Violations
  - Entity: if the primary key value is null (<Jay,Smith,null,60/04/05,M,28000,null,7>)
  - Referential Integrity: if the value of a foreign key refers to a tuple that does not exist (<Jay,Smith,999887777,60/04/05,M,28000,null,7>)
- Possible DBMS Actions on Violations
  - reject, cascade, modify, inform

Delete Operations

- Delete Operations
  - delete a tuple(s) from a relation
  - can violate referential integrity only
  - delete the WORKS_ON tuple with SSN=`123456789' and PNO=1 (acceptable)
  - delete the EMPLOYEE tuple with SSN=`333445555‘ (violates the constraint)

Delete Operations

- Delete the WORKS_ON tuple with ESSN='999887777' and PNO=10

This deletion is acceptable

- Delete the EMPLOYEE tuple with SSN='999887777'

This deletion is not acceptable, because tuples in WORKS_ON refer to this tuple. Hence, if the tuple is deleted, referntial integrity violations will result

Delete Operations

- Delete the EMPLOYEE tuple with SSN='333445555'.

  This deletion will result in even worse referential integrity violations, because the tuple involved is referenced by tuples from the EMPLOYEE, DEPARTMENT, WORKS_ON and DEPENDENT relations.

Update Operations

- Modify Operations
  - modify the value of an attribute
  - can violate all four relational constraints
  - update the DNO of the EMPLOYEE tuple with SSN=`999887777' to 7 (referential integrity)
  - update the SSN of the EMPLOYEE tuple with SSN=`999887777' to `987654321` (key and referential integrity violated)

Update Operations

- Update the SALARY of the EMPLOYEE tuple with SSN='999887777' to 28000.

  Acceptable

- Update the DNO of the EMPLOYEE tuple with SSN='999887777' to 1.

  Acceptable

Update Operations

- Update the DNO of the EMPLOYEE tuple with SSN='999887777' to 7.

Unacceptable, because it violates referential integrity.

- Update the SSN of the EMPLOYEE tuple with SSN='999887777' to '987654321'.

Unacceptable, because it violates primary key and referential integrity constraints.